Moral Tensors and DecisionProofs

Compiling Language into an Auditable, Grounded Safety Layer

Structure-preserving representation before decision contraction.

`pip install erisml-compiler` · `github.com/ahb-sjsu/erisml-compiler · erisml-lib · agi-hpc`

Andrew H. Bond

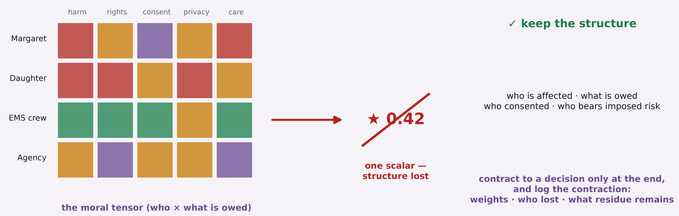
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PyTorch Conference North America 2026 — Responsible AI poster

1

Don't collapse to a scalar.



the moral tensor (who × what is owed)

✓ keep the structure

who is affected · what is owed
who consented · who bears imposed risk

contract to a decision only at the end, and log the contraction:
weights · who lost · what residue remains

2

pip install erisml-compiler — structure before the scalar



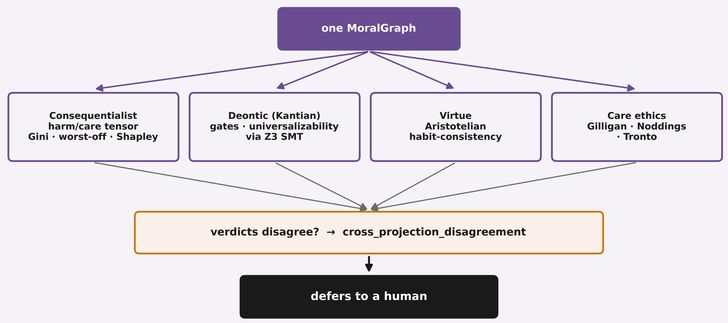
MoralGraph nodes: stakeholder · act · maxim · commitment · fact · norm — carries a canonical SHA-256 hash

extractor tiers: rule (deterministic, silicon-castable) · LLM (NRP / vLLM + critic) · probe (LaBSE head)

alpha v0.9.0 · 477 tests · MIT · Zenodo DOI 10.5281/zenodo.20659432

3

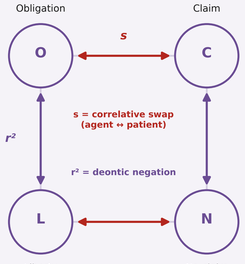
Four lenses. No silent winner.



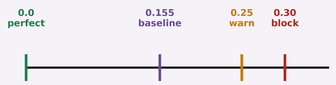
"We run four and show you where they conflict — we don't pick one and hide it."

4

Is this 'good'? → Does it preserve the bonds?



s and r² commute → V4 (Klein four-group, order 4) — measured



Bond Index: does the judgment survive the swap?

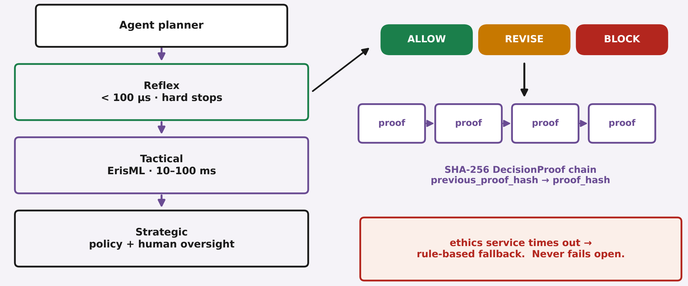
multi-rank tensors: party × time × action × coalition

V4 claim machine-checked: Lean 4 + Mathlib (formal/HohfeldV4.lean)

full D4 (order 8) posited, now tested: a 09/26 hunt in a small LM found no quarter-turn (non-abelian signature absent). V4 stands.

5

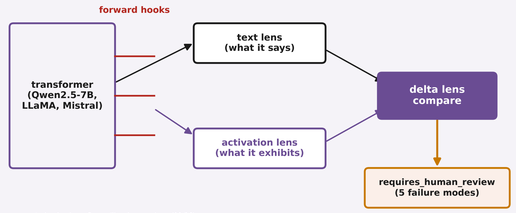
Safety in the loop. Fails safe.



between the agent's planner and its actuators

6

What the model says vs what it exhibits



Activation lens: first calibration numbers (09/26). Consumer-metric gating catches 19-21 pp more real behavior changes than raw L2 at matched flag rates. Replicated 2/2: second model + second transform. Probe mid-layers: the final-layer β map fails held-out validation in all three runs.

turboquant-pro (PyPI): PyTorch-native compression
HF 1-liner: TurboQuantCache() · Triton + Volta kernels · vLLM ~5× KV
live here: 3-bit codec on NATS · claims CI-gated · measured false-clear diag (v2.0.0a3)

7

Replay the judgment, and the regime that made it

full_response	NORMAL	EMERGENCY
Margaret	0.185 forbid	0.705 prefer
Daughter	0.402 neutral	0.902 prefer
EMS crew	0.318 forbid	0.902 prefer
Agency	0.200 forbid	0.802 prefer

DEME's scores: stand_by 0.834 → full_response 0.804, and stand_by inverts to 0.307

elevation: detector 0.97 ≥ 0.90 → GRANTED, paramedics only, auto-reverts, both edges audited

conf 0.89 → REFUSED: human review and the EMS call, no record share

worst-off: Daughter → Margaret
Shapley authority: Margaret → EMS

Three DecisionProofs: normal, emergency, the elevation edge

$ python -m erisml.examples.care_robot_regimes reproduces every number

In the robot, DEME is a firewall: it vets each proposed act, never picks one



any stop: robot holds

Change one input, nothing else:	base	one changed
fire call: one of two detectors silent	passed	stopped: governor
ambulance: her refusal heard / not	stopped: DEME	passed
help her up: she asked / did not	passed	stopped: DEME

9 pairs, each prediction committed before its run; 2 of 18 wrong, both explained in the record

Found by this test, then fixed:
the lift she had not asked for passed DEME until 18 Oct.
Rerun: only that verdict changed.

Simulated 3D home, scripted proposals, no hardware

1. Structure before contraction — keep the tensor, log the collapse + the residue.
2. Pluralism is the responsible move — four lenses, defer to a human on conflict.
3. Auditable by construction; fails safe, never open.

alpha v0.9.0 · text path solid · activation lens early; first calibration 09/26 (consumer-metric, replicated 2/2) · ranks 1–3 real, higher partial
V4 measured, D4 posited (Lean + Mathlib verified)
embodied = simulated 3D home, scripted, no hardware · agents = live

companion paper · **erismL-compiler** · **erismL-lib** · **agi-hpc**